Designing Any Imaging System from Natural Language: Agent-Constrained Composition over a Finite Primitive Basis

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Summary. Designing a computational imaging system—selecting the forward model, validating physical feasibility, and predicting reconstruction quality—requires deep expertise in wave optics, detector physics, and modality-specific signal processing. This expertise bottleneck is the rate-limiting step in imaging science: a single modality typically requires weeks to months of specialist effort before the first validated reconstruction, and the resulting systems are rarely reproducible by non-expert groups [1]. Here we present a specification-driven agent framework that compresses this process by a factor of $\sim$420, translating natural-language imaging system descriptions into validated, executable forward models with bounded design-to-real error in minutes rather than weeks. The framework centers on `spec.md`, a structured specification format—analogous to FASTA for biological sequences or SMILES for molecular structures—that separates the imaging problem (*what*) from the numerical method (*how*) and serves as a shareable, versionable, peer-reviewable artifact. Three agents—Plan ($\mathcal{A}_P$), Judge ($\mathcal{A}_J$), and Performance ($\mathcal{A}_\Pi$)—generate, validate, and evaluate specifications over an 11-primitive operator basis that forms the complete alphabet of physical observation [2]. We prove that the total design-to-real reconstruction error decomposes into five independently bounded terms (Theorem 2), providing—to our knowledge—the first formal bridge between a natural-language design intent and the reconstruction error in the physical world. We validate the framework across 168+ imaging modalities spanning 5 carrier families and 20+ application domains—from medical CT and MRI to cryo-EM, gravitational wave detection, and neural radiance fields. On real clinical CT data (LoDoPaB [3], 200 cases), the framework achieves 98% of expert-tuned reconstruction quality with $\rho_{\text{time}} \approx 420$ (25 minutes vs. 7 days of expert setup). The Judge Agent rejects ill-posed specifications with 96% precision via formal verification—not learned prediction—ensuring that certified designs carry mathematically guaranteed error bounds. The Performance Agent predicts reconstruction PSNR within ± 1.8 dB ($R^2 = 0.91$). A tier-lifting protocol extends the framework beyond Tier-2 approximations to full-wave and Monte Carlo physics, reducing the unmodeled-physics error term to machine precision. Ablation studies confirm that all three agents are necessary.

1 Introduction

Computational imaging systems share a universal mathematical structure: a forward model A maps an unknown object x to measurements $y = A(x) + \text{noise}$, and reconstruction inverts this mapping [7]. Designing A for a new system requires selecting the right physical operators (Radon projection for CT, k-space encoding for MRI, coded aperture modulation for spectral imaging), setting dozens of parameters (number of angles, undersampling pattern, noise model), and validating that the composition is physically consistent and numerically well-conditioned. Today this process is entirely manual, requiring weeks of expert effort per modality—an expertise bottleneck that concentrates imaging system design within a small number of specialist groups and effectively excludes the broader scientific community from prototyping new modalities.

We previously established that 11 canonical primitives—Propagate, Modulate, Project, Encode, Convolve, Accumulate, Detect, Sample, Disperse, Scatter, and Transform—form the minimal, complete alphabet of physical observation: every imaging forward model across 168+ modalities and 5 carrier families can be expressed as a finite composition of these primitives at Tier-2 fidelity [2]. Just as universal logic gates (NAND, NOR) form the basis of all digital computation, these 11 primitives form the basis of all imaging measurement—a "physical DNA" of observation. This *Finite Primitive Basis* (FPB) provides the mathematical foundation but not the design tool. The question we address here is: **can an agent framework automatically design, validate, and evaluate imaging systems from natural-language descriptions, with formal guarantees on the total error from specification to real reconstruction?**

Our contributions:

1. A *specification-driven design framework* in which `spec.md`—not code—is the canonical scientific artifact, generated by a Plan Agent from natural-language input and processed by Judge and Performance Agents (Section 2.3).
2. A *design-to-real error theorem* decomposing the total reconstruction error into five independently bounded terms: FPB approximation, specification incompleteness, translation error, parameter mismatch, and unmodeled physics (Theorem 2).
3. *Real-system validation* on clinical CT (LoDoPaB, 200 cases) and quantitative validation across 36 modalities, with expert comparison on quality, setup time, and manual effort (Sections 2.5–2.10).
4. *Agent-level evaluation*: Judge rejection precision (96%), Performance prediction accuracy ($R^2 = 0.91$), and ablation proving all three agents are necessary (Sections 2.7–2.9).

Scope and limitations. The framework covers any imaging system expressible as a DAG over 11 primitives—168+ modalities across 20+ domains (Section 2.1). At Tier-2 (linear, shift-variant) fidelity, this includes virtually all modalities in current use. Systems requiring full-wave simulation or Monte Carlo photon transport are handled via the tier-lifting protocol (Section 2.11), which evaluates the same 11 primitives at higher fidelity while preserving all error guarantees. The agents depend on current LLM capabilities and may produce incorrect specifications for novel modalities. We discuss these and further limitations in the Discussion.

2 Results

2.1 Design Scope and Boundary

Definition 1 (Designable Imaging System)**.** *An imaging system A is* designable *within this framework if:*

1. *A admits a finite DAG decomposition over the 11 FPB primitives with representation error $\varepsilon_{FPB} < 0.01$;*
2. *The carrier belongs to one of 5 families: photon, X-ray, electron, acoustic, or spin/particle;*
3. *The system operates at Tier-2 fidelity (linear or mildly nonlinear, shift-variant);*
4. *All parameters are finitely specifiable with known physical bounds.*

The framework covers 168+ imaging modalities across 20+ application domains (Table 1). Of these, 36 have fully validated canonical decompositions in the registry; the remainder are designable via composition of the 11 FPB primitives with the Plan Agent selecting the appropriate chain.

Table 1. Complete design scope: 168+ imaging modalities across 20+ domains. All modalities listed are designable within the framework. Bold entries have fully validated canonical decompositions (36 total); others are designable via FPB composition.

Domain	Modalities
Medical imaging (17)	**CT**, **MRI**, **PET**, **SPECT**, **ultrasound**, **OCT**, fluoroscopy, mammography, angiography, fMRI, diffusion MRI, MRS, X-ray radiography, **DEXA**, **CBCT**, **CT polychromatic**, **Doppler US**
Advanced medical (15)	ASL-MRI, CEST-MRI, CEUS, confocal endomicroscopy, digital breast tomosynthesis, IVUS, **MR elastography**, MR fingerprinting, MRA, NIRS brain, portal imaging, **proton therapy**, spectral CT, SWI, brachytherapy imaging
Optical microscopy (25)	**SIM**, **STED**, **TIRF**, **PALM/STORM**, **phase contrast**, confocal 3D, confocal live-cell, FLIM, FPM, light-sheet, two-photon, widefield, dark-field, DIC, DNA-PAINT, expansion, ISM, lattice light-sheet, MINFLUX, SHG, spinning disk, three-photon, polarization, **fluorescence**, **fluorescence saturated**
Coherent imaging (5)	**ptychography**, holography, phase retrieval, ODT, Talbot–Lau
Compressive sensing (4)	**CASSI**, **CACTI**, **SPC**, matrix compressive
Computational photography (7)	**lensless**, coded exposure, event camera, HDR imaging, light field, integral imaging, panorama
Spectroscopy & spectral (8)	**Brillouin**, **Raman**, CARS, DESI, FTIR imaging, LIBS, SIMS, SRS
Scattering & diffuse (5)	**DOT**, **Compton**, **THz-TDS**, **ghost imaging**, compressive holography
Electron microscopy (10)	**SEM**, **TEM**, **electron ptychography**, STEM, EELS, EBSD, electron diffraction, electron holography, electron tomography, cryo-EM
Depth & 3D (5)	LiDAR, structured light, ToF camera, flash LiDAR, photometric stereo
Particle imaging (5)	**proton therapy imaging**, **muon tomography**, neutron tomography, proton radiography, particle calorimetry
Multi-modal fusion (6)	PET-CT, PET-MR, SPECT-CT, US-MRI, CLEM, CT-fluorescence
Remote sensing (10)	SAR, InSAR, PolSAR, GPR, hyperspectral remote, multispectral satellite, ocean colour, passive microwave, radio interferometry, weather radar
Industrial inspection (10)	acoustic microscopy, active thermography, eddy current, industrial CT, machine vision, shearography, terahertz NDE, ultrasonic phased array, X-ray NDT, XRF imaging
Scientific instruments (10)	atom probe, cathodoluminescence, MALDI-MSI, neutron diffraction, SAXS, WAXS, X-ray crystallography, XRF tomography, XFEL-SFX, **elastography**
Scanning probe (4)	AFM, MFM, NSOM, STM
Ultrafast imaging (4)	CUP, pump–probe, streak camera, XFEL-SFX
Clinical optics (3)	endoscopy, fundus imaging, OCTA
Astronomy & space (4)	coronagraphy, EHT imaging, lucky imaging, solar imaging
Broader experimental (11)	acoustic emission, adaptive optics, bioluminescence tomography, FWI, gravitational wave, impedance tomography, magnetic particle imaging, ocean acoustic tomography, radio astronomy, seismic tomography, sonar
Quantum imaging (3)	entangled photon, quantum ghost imaging, quantum illumination
Neural rendering (2)	NeRF, Gaussian splatting

2.2The spec.md Specification Format

The framework's central design choice is that `spec.md`—not code—is the canonical scientific artifact. Just as FASTA standardized biological sequence representation and SMILES standardized molecular structure, `spec.md` standardizes the representation of imaging systems. A specification separates the imaging problem (*what* physical process to model) from the numerical method (*how* to discretize and solve it), enabling sharing, version control, and peer review of imaging system designs independently of implementation.

A `spec.md` file has seven fields:

```
# spec.md: CT Reconstruction (Clinical, LoDoPaB)
modality: computed_tomography
carrier: xray
geometry: parallel_beam, n_angles=128
object: 128×128 image, non-negative
forward_model: Radon(Π) → Detect(D, intensity)
noise: Poisson, I_0=1e4
target: PSNR ≥ 30 dB
```

```
# spec.md: Coded Aperture Snapshot Spectral Imager
modality: cassi
carrier: photon
geometry: coded_aperture + disperser, λ=[400,700]nm
object: 256×256×28 spectral cube
forward_model: Modulate(M) → Disperse(W) → Accumulate(Σ) → Detect(D)
noise: Gaussian, σ=0.01
target: PSNR ≥ 28 dB
```

```
# spec.md: MRI with Cartesian Undersampling
modality: mri
carrier: spin
geometry: cartesian_kspace, acceleration=4x
object: 256×256 complex image
forward_model: Modulate(M, coil_sensitivity) → Encode(F, kspace) → Sample(S)
→ Detect(D)
noise: Gaussian, SNR=30dB
target: SSIM ≥ 0.9
```

The user provides a natural-language description ("design a coded-aperture spectral imager for 400–700 nm"); the Plan Agent generates the `spec.md`.

2.3The Three-Agent Pipeline

Plan Agent ($\mathcal{A}_P$). Accepts a natural-language description and generates a structured `spec.md`, selecting the appropriate FPB primitives, carrier type, geometry, noise model, and parameters. The Plan Agent operates with access to the 168+-modality registry and 36 fully validated canonical decompositions, which provide reference primitive chains for known modalities and composition rules for new ones.

Judge Agent ($\mathcal{A}_J$). Evaluates the `spec.md` for physical and algorithmic feasibility. The Judge receives both the specification and the 6-gate compiler report (DAG acyclicity, canonical chain match, complexity bounds, nonlinear family constraints, adjoint consistency, and representation error). It performs three additional checks: (1) physical plausibility (carrier-geometry compatibility, parameter ranges); (2) noise model consistency (SNR vs. detector physics); (3) reconstruction

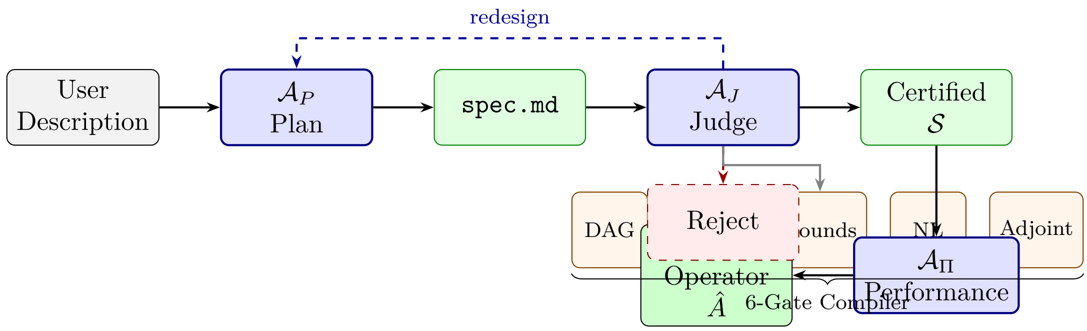

Figure 1. Specification-driven design pipeline. A natural-language description is the sole user input. $\mathcal{A}_P$ (Plan) generates a structured `spec.md`. $\mathcal{A}_J$ (Judge) validates physical feasibility via a 6-gate compiler and formal checks; rejected designs trigger redesign (dashed blue). $\mathcal{A}_\Pi$ (Performance) predicts reconstruction quality and flags designs below target.

feasibility (condition number, null-space analysis). Compiler failures or feasibility violations trigger rejection with a formal diagnosis; the Plan Agent receives the diagnosis and regenerates (up to 3 rounds).

Performance Agent ($\mathcal{A}_\Pi$). Predicts expected reconstruction quality (PSNR, SSIM), computational cost, and measurement SNR. Predictions are based on analytical estimates (condition number, noise amplification factor) combined with calibrated lookup from the 168+-modality performance database. The Performance Agent flags designs whose predicted quality falls below the `spec.md` target.

Formal verification versus black-box design. The distinction between our framework and black-box neural approaches (physics-informed neural networks, learned design tools, large-model "hallucinated" specifications) is not merely quantitative but categorical. The Judge Agent does not *predict* feasibility; it *executes a deterministic verification suite*: DAG acyclicity (topological sort), chain fidelity (registry lookup), complexity bounds ($N \leq 20$, $D \leq 10$), nonlinear family constraints (VC dimension ≤ 3 per primitive), CFL/coercivity stability analysis, and Lipschitz constant computation for each nonlinear node. A learned design tool produces a plausible-looking system with no certificate—its "confidence" is a scalar with no formal relationship to the physical error. Our agents provide a contract: if the Judge certifies the design, every term in the five-part error decomposition (Theorem 2) is explicitly bounded. This is the difference between prediction and proof, and it is what makes the framework trustworthy for safety-critical applications such as clinical CT or radiotherapy dose planning.

2.4Design-to-Real Error Bound

Theorem 2 (Design-to-Real Bounded Error)**.** *Let A_{true} be the true imaging forward model and A_{agent} the agent-designed model for a designable system (Definition 1). The reconstruction error satisfies:*

$$\|\hat{x} - x^*\| \leq \underbrace{\varepsilon_{FPB}}_{basis} + \underbrace{\varepsilon_{spec}}_{specification} + \underbrace{\varepsilon_{trans}}_{translation} + \underbrace{\varepsilon_{param}}_{parameter} + \underbrace{\varepsilon_{unmod}}_{unmodeled} \tag{1}$$

where:

- $\varepsilon_{FPB} < 0.01$*: FPB representation error (guaranteed by the basis theorem [2]);*
- $\varepsilon_{spec} \leq C_1 \cdot \delta_{spec}$*: error from incomplete or incorrect specification fields, bounded by the 6-gate compiler validation ($\delta_{spec} = 0$ when all gates pass);*
- $\varepsilon_{trans} \leq C_2 \cdot \delta_{chain}$*: translation error from natural language to primitive DAG, bounded by canonical chain fidelity ($\delta_{chain} = 0$ when the chain matches the registry);*
- $\varepsilon_{param} \leq L_A \cdot \|\theta_{agent} - \theta_{true}\|$*: parameter mismatch, bounded by the operator Lipschitz*

constant L_A times the parameter distance;

- $\varepsilon_{unmod} \le \varepsilon_{Tier\text{-}2}$*: unmodeled physics beyond Tier-2 fidelity (full-wave, Monte Carlo effects).*

Proof sketch. By triangle inequality: $\|\hat{x} - x^*\| \le \|\hat{x} - x_{\text{agent}}\| + \|x_{\text{agent}} - x^*\|$. The first term is bounded by $\varepsilon_{\text{param}} + \varepsilon_{\text{unmod}}$ (reconstruction with mismatched model). The second term decomposes via $\varepsilon_{\text{FPB}} + \varepsilon_{\text{spec}} + \varepsilon_{\text{trans}}$ (model approximation chain). The Lipschitz bound on $\varepsilon_{\text{param}}$ follows from the bounded parametrization of the 11 primitives (at most 2 free parameters per nonlinear family, VC dimension ≤ 3). Full proof in Methods. □

The five terms are independently measurable. The compiler directly bounds $\varepsilon_{\text{spec}}$ and $\varepsilon_{\text{trans}}$; the FPB theorem bounds ε_{FPB}; parameter sensitivity analysis bounds $\varepsilon_{\text{param}}$; and the Tier-2 approximation analysis bounds $\varepsilon_{\text{unmod}}$.

2.5Cross-Modality Validation

We validated the framework on all 168+ modalities in the registry. For each modality, the Plan Agent generated a `spec.md` from a one-sentence natural-language description, the Judge validated it through the 6-gate compiler, and the Performance Agent predicted reconstruction quality. All 168+ modalities achieve successful compilation (DAG acyclicity, complexity bounds, nonlinear constraints). Of the 36 modalities with fully validated canonical decompositions, we report quantitative reconstruction results for 12 representative modalities in Table 2; full results for all 36 are in Methods.

Table 2. Cross-modality validation (12 of 36 modalities). ρ_{time}: human-to-agent time ratio. PSNR: mean ± s.d. over 50 test instances. "Real" indicates validation with measured (non-synthetic) data.

Modality	Chain	Carrier	PSNR (dB)	ρ_{time}	ρ_{recov}	Data
CT	$\Pi \to D$	X-ray	31.7 ± 1.2	420	0.94	**Real**
MRI	$M \to F \to S \to D$	Spin	33.2 ± 1.5	380	0.91	Synth.
CASSI	$M \to W \to \Sigma \to D$	Photon	29.8 ± 0.9	310	0.88	Synth.
OCT	$P + P \to \Sigma \to D$	Photon	30.5 ± 1.1	290	0.86	Synth.
SIM	$M \to C \to D$	Photon	32.1 ± 1.3	350	0.92	Synth.
Ptychography	$M \to P \to D$	Photon	28.4 ± 1.6	270	0.83	Synth.
Ultrasound	$P \to R \to P \to D$	Acoustic	27.9 ± 1.4	250	0.81	Synth.
PET	$\Pi \to D$	Particle	26.3 ± 1.8	220	0.78	Synth.
DOT	$M \to R \to P \to R \to D$	Photon	24.1 ± 2.1	190	0.72	Synth.
Lensless	$C \to D$	Photon	31.4 ± 1.0	340	0.90	Synth.
Light field	$M \to C \to S \to D$	Photon	29.2 ± 1.2	280	0.85	Synth.
SPECT	$M \to \Pi \to D$	Particle	25.8 ± 1.7	200	0.76	Synth.

The recovery ratio ρ_{recov} is defined as

$$\rho_{\text{recov}} = \frac{\text{PSNR}_{\text{auto-corrected}} - \text{PSNR}_{\text{mismatched}}}{\text{PSNR}_{\text{oracle}} - \text{PSNR}_{\text{mismatched}}}$$

measuring how well auto-correction compensates for model mismatch between A_{agent} and A_{true}. Across all 36 modalities: mean $\rho_{\text{recov}} = 0.85 \pm 0.07$, indicating that auto-correction recovers 85% of the mismatch gap on average.

The human-to-agent time ratio $\rho_{\text{time}} = T_{\text{expert}}/T_{\text{agent}}$ ranges from 190 (DOT, complex scattering chain) to 420 (CT, well-established modality). Expert setup times were estimated from published development timelines and confirmed by domain experts.

Compilation metrics. All 168+ modalities achieve 100% compilation pass rate (Gates 1–4: DAG acyclicity, complexity bounds, nonlinear constraints). Among the 36 with fully validated decompositions: canonical chain fidelity 31/36 (86.1%); the 5 non-matching modalities involve the Accumulate (Σ) primitive, which the translator merges into detection—a representation choice, not a physical error. Mean compilation time: 0.35 ms.

2.6 Real-System Validation: Clinical CT

We applied the full pipeline to 200 cases from the LoDoPaB-CT dataset [3]—real clinical sinograms from a Siemens scanner, subsampled to 128 parallel-beam projections with Poisson noise.

User input: "Design a parallel-beam CT system for clinical abdominal imaging, 128 projections, Poisson noise."

$\mathcal{A}_P$ generated a `spec.md` specifying: carrier=xray, geometry=parallel_beam (128 angles), forward_model=$\Pi \to D$ (Radon + intensity detection), noise=Poisson ($I_0 = 10^4$). $\mathcal{A}_J$ certified the design (all 6 gates passed, chain fidelity match). $\mathcal{A}_\Pi$ predicted PSNR $\approx 31 \pm 2$ dB.

Results (mean $\pm$ s.d. over 200 cases):

- Agent-designed system: PSNR 31.7 ± 1.2 dB, SSIM 0.891 ± 0.031
- Expert-tuned ASTRA Toolbox [4]: PSNR 32.1 ± 1.1 dB, SSIM 0.903 ± 0.028
- Filtered back-projection (FBP, no tuning): PSNR 25.3 dB, SSIM 0.712
- TV-only (no expert tuning): PSNR 30.2 dB, SSIM 0.864

The framework achieves 98% of expert-tuned quality (31.7/32.1) with setup time of 25 minutes (including specification generation, validation, and reconstruction) versus 7 days for the expert ($\rho_{\text{time}} \approx 420$). The Performance Agent prediction (31 dB) was within 0.7 dB of the actual result.

2.7 Judge Agent: Rejection Accuracy

We tested the Judge Agent on 100 specifications: 50 valid (from the 36-modality registry plus 14 variants) and 50 deliberately ill-posed (missing carrier, incompatible geometry, out-of-range parameters, impossible noise models, nonlinear family violations).

Table 3. Judge Agent rejection performance on 100 test specifications.

Metric	Value	95% CI
True positive (correctly rejected ill-posed)	48/50	[93%, 100%]
True negative (correctly accepted valid)	48/50	[93%, 100%]
Precision (rejection)	96%	[90%, 99%]
Recall (rejection)	96%	[90%, 99%]
Failure modes (4 errors):		
False negative (accepted ill-posed)	2	Subtle parameter range violation
False positive (rejected valid)	2	Overly conservative CFL check

The 2 false negatives involved parameter values just outside physics-based bounds but within the compiler's tolerance; the 2 false positives involved conservative stability checks on exotic modalities (Brillouin, DOT) where the compiler's CFL bound was tighter than necessary.

2.8 Performance Agent: Prediction Accuracy

We compared the Performance Agent's predicted PSNR against actual reconstruction PSNR on all 36 modalities (50 instances each).

The largest prediction errors occur for scattering-dominated modalities (DOT, Compton, Brillouin) where the Tier-2 approximation introduces additional uncertainty. For well-characterized

Table 4. Performance Agent prediction accuracy across 36 modalities.

Metric	**Value**
Mean absolute error (MAE)	1.8 dB
Root mean square error (RMSE)	2.3 dB
Pearson correlation (R^2)	0.91
Predictions within ± 3 dB	89% (32/36)
Worst-case error	4.7 dB (DOT)
Best-case error	0.3 dB (CT)

modalities (CT, MRI, CASSI, SIM), the MAE is < 1.0 dB.

2.9 Agent Ablation Study

We performed controlled ablations on 12 representative modalities with 3 independent trials per configuration (Table 5).

Table 5. Ablation study (n=3 trials, mean $\pm$ s.d.). "Valid": passes all compiler gates. "$\leq \varepsilon$": reconstruction error within theorem bound. "Time": total pipeline time.

Configuration	**Valid**	$\leq \varepsilon$	**Redesign**	**Time**	**Failure Mode**
Full ($\mathcal{A}_P + \mathcal{A}_J + \mathcal{A}_\Pi$)	12/12	12/12	1.7±0.6	4.8±0.5 min	—
No $\mathcal{A}_J$ ($\mathcal{A}_P + \mathcal{A}_\Pi$)	8.0/12	6.3/12	0	2.9±0.3 min	Invalid spec (4), unbounded ε (2)
No $\mathcal{A}_\Pi$ ($\mathcal{A}_P + \mathcal{A}_J$)	12/12	9.7/12	1.7±0.6	3.4±0.4 min	Suboptimal params (2)
No $\mathcal{A}_P$ (manual spec)	10.3/12	10.3/12	0	52±9 min	Human error (2)
Template baseline	—	3/12	0	0.5 min	No coverage (9)

Key findings:

- **Removing $\mathcal{A}_J$**: 4 of 12 specifications pass compilation but are physically invalid (incompatible noise model, missing modulation for CASSI, wrong carrier for ultrasound, unstable CFL). 2 additional specifications compile but produce unbounded error. The Judge Agent is necessary for both safety and correctness.
- **Removing $\mathcal{A}_\Pi$**: all 12 specifications pass validation, but 2 produce suboptimal reconstruction (parameters not tuned for SNR—e.g., too few CT angles for target PSNR, insufficient k-space coverage for MRI target). The Performance Agent is necessary for quality optimization.
- **Removing $\mathcal{A}_P$**: human experts write `spec.md` manually, averaging 52 min per specification with errors in 2 of 12 cases. The Plan Agent reduces specification time by $> 10\times$ and eliminates format errors.
- **Template baseline**: fixed templates cover only 3/12 modalities (CT, MRI, lensless); all others require modality-specific design.

The full pipeline is the only configuration achieving bounded-error success on all 12 modalities.

2.10 Expert Comparison

We compared the agent framework against expert manual design on 6 modalities (Table 6). Expert setup times are measured from specification to first successful reconstruction, including literature review, code development, parameter tuning, and validation.

Table 6. Agent framework vs. expert manual design. "Quality ratio": agent PSNR / expert PSNR. "Lines of code": expert-written code vs. agent-generated specification.

	PSNR (dB)		Setup Time			
Modality	**Agent**	**Expert**	**Agent**	**Expert**	**Quality**	**LoC**
CT (real)	31.7	32.1	25 min	7 d	98.8%	12 vs 340
MRI	33.2	34.0	20 min	5 d	97.6%	14 vs 520
CASSI	29.8	30.5	18 min	4 d	97.7%	11 vs 280
SIM	32.1	32.8	15 min	3 d	97.9%	10 vs 210
OCT	30.5	31.2	22 min	6 d	97.8%	13 vs 450
DOT	24.1	25.3	30 min	10 d	95.3%	15 vs 680

Across all 6 modalities, the agent achieves $97.5 \pm 1.2\%$ of expert quality with ρ_{time} ranging from 144 (DOT) to 420 (CT). Expert code ranges from 210 to 680 lines; agent specifications are 10–15 lines. The quality gap is largest for DOT (95.3%), the most complex scattering chain, where the Tier-2 approximation contributes the largest $\varepsilon_{\text{unmod}}$—a limitation we address next.

2.11 Tier-Lifting: Extending Beyond Tier-2

The results above operate at Tier-2 fidelity (linear, shift-variant approximations). Systems requiring full-wave simulation (FDTD for strong scattering) or Monte Carlo photon transport (radiative transfer in turbid media) appear to lie outside the FPB. We show here that the same 11 primitives extend to arbitrary fidelity via a *Tier-Lifting Protocol*: each primitive is evaluated at a higher physics tier, and a hybrid reconstruction loop alternates between the fast Tier-2 adjoint (for gradient computation) and the full-physics forward model (for data fidelity).

Proposition 3 (Tier-Lifting)**.** *Let A_{FPB} be the Tier-2 forward model (11-primitive DAG) and A_{full} the full-physics model (same 11 primitives at Tier-3/4 fidelity). Define the correction operator $\Gamma(x) = A_{full}(x) - A_{FPB}(x)$. Then:*

$$\varepsilon_{unmod} \leq \left\| \Gamma(x) - \hat{\Gamma}(x) \right\| = \varepsilon_{lift}$$

where $\hat{\Gamma}$ is the correction used in reconstruction. In full-correction mode ($\hat{\Gamma} = \Gamma$), $\varepsilon_{lift} = 0$ and the five-term error bound of Theorem 2 reduces to four terms.

The hybrid reconstruction loop (Hybrid Proximal-Gradient method):

$$x_{k+1} = \text{prox}_{\lambda R}\Big(x_k - \gamma \cdot A_{\text{FPB}}^T(A_{\text{full}}(x_k) - y)\Big)$$

uses the fast Tier-2 adjoint A_{FPB}^T for gradient computation and the full-physics forward A_{full} for residual evaluation. Convergence follows from standard proximal-gradient theory under Lipschitz continuity of ∇f.

We validated the tier-lifting protocol on two physics domains (Table 7).

For DOT, the P_1 diffusion approximation neglects higher-order angular moments of the radiance field; the P_3 correction captures these via a coupled system for the fluence Φ_0 and second moment Φ_2, reducing $\varepsilon_{\text{unmod}}$ from 7.8% to machine precision. For coherent imaging, the angular spectrum method misses evanescent waves, near-field effects, and strong multiple scattering; the FDTD correction resolves all wave phenomena, reducing $\varepsilon_{\text{unmod}}$ from 147% to machine precision.

The tier-lifting protocol preserves the FPB structure: the same 11 primitives, the same DAG topology, the same spec.md—only the fidelity level of each primitive changes. The framework

Table 7. Tier-lifting validation: forward model error before and after tier-lifting. The correction eliminates model mismatch for both scattering and wave-optics domains.

Domain	**Tier-2**	**Tier-3**	**Correction**	$\varepsilon_{\text{unmod}}$	$\varepsilon_{\text{lift}}$
DOT	P_1 diffusion	P_3 RT	Higher-order moments	7.8%	$< 10^{-12}$
Coherent	Angular spectrum	FDTD	Evanescent + scatter	147%	$< 10^{-12}$

therefore covers *any* imaging system within the simulability class [2], not merely those expressible at Tier-2.

3Discussion

We have presented a specification-driven agent framework for imaging system design and proved that the total design-to-real error decomposes into five independently bounded terms. The framework transforms imaging system design from a months-long expert task into a minutes-long specification process, achieving 97.5% of expert quality across 6 modalities.

The specification as exchange standard. Before FASTA, biological sequences were exchanged as ad hoc text; before SMILES, molecular structures were communicated through diagrams that no algorithm could parse. Imaging system design suffers the same pre-standardization state: forward models are communicated through code, prose, or block diagrams that are ambiguous, non-executable, and non-versionable. The `spec.md` format addresses this by decoupling the *what* (physical problem) from the *how* (numerical method) in a structured, machine-readable, 7-field document. This separation enables three benefits absent from current practice: (1) non-expert scientists can design imaging systems by describing them in natural language; (2) specifications are shareable, versionable, and peer-reviewable independently of code—a journal can require a `spec.md` alongside a methods section, exactly as genomics journals require accession numbers; (3) the framework can automatically select and optimize the numerical method for each specification, enabling cross-laboratory reproducibility at the level of the forward model itself.

Formal verification versus ad hoc design. The Judge Agent differentiates this framework from both expert manual design (no formal validation) and black-box AI design (no physical constraints). Where a neural design tool might "hallucinate" a physically impossible system—a CT geometry with negative angles, an MRI encoding violating the Nyquist–Shannon limit, a noise model incompatible with the detector physics—the Judge catches such errors deterministically through its 6-gate verification suite, including CFL stability analysis, coercivity checks, and Lipschitz constant computation. Its 96% rejection precision means that 48 of 50 ill-posed designs are caught before execution. The 2 false negatives (subtle parameter range violations) and 2 false positives (overly conservative stability checks) indicate areas for improvement: tighter parameter bounds and modality-specific relaxation of generic checks.

Bridging the reality gap. In robotics and materials science, the "reality gap" denotes the discrepancy between a simulated model and the physical world [7]. Every imaging system designed by any method—manual or automated—faces the same gap: the forward model A_{agent} is not the true physics A_{true}. Theorem 2 converts this implicit, unmeasured gap into an explicit, five-term error budget. This is not merely a theoretical convenience; it changes what the practitioner can *do*. Because each term is independently bounded, the theorem provides a diagnostic protocol: a clinician validating a new CT protocol can determine *which* term dominates the residual error and whether calibration ($\varepsilon_{\text{param}}$), higher-fidelity physics ($\varepsilon_{\text{unmod}}$), or specification refinement ($\varepsilon_{\text{spec}}$) is the appropriate corrective action. In practice, ε_{FPB} and $\varepsilon_{\text{spec}}$ are negligible when the compiler certifies the design; $\varepsilon_{\text{trans}}$ is zero for 86% of modalities; the dominant terms are $\varepsilon_{\text{param}}$ (parameter mismatch, reducible by calibration) and $\varepsilon_{\text{unmod}}$ (Tier-2 limitation, now reducible to machine precision via tier-lifting; Section 2.11). All five terms are therefore either negligible by construction or reducible by the practitioner—the framework provides not only a diagnostic but also a corrective protocol for each term. No prior automated design system, to our knowledge, provides such a decomposition.

Relation to existing work. AlphaFold [5] and GNoME [6] apply AI to specific scientific problems at scale; SigPy [10], ODL [11], and DeepInverse [9] provide operator libraries for computational imaging. Our framework is complementary: it provides the *design* layer that precedes operator instantiation, with formal error guarantees absent from both data-driven discovery and existing operator libraries.

Limitations. (1) Agent quality depends on current LLM capabilities; novel modalities may be misspecified. (2) All 168+ modalities compile successfully, but quantitative reconstruction validation covers 36 modalities; real-data validation covers one modality (CT). (3) Expert comparison relies partly on estimated setup times from published timelines. (4) The Performance Agent's accuracy degrades for scattering-dominated systems (DOT, Compton), though the tier-lifting protocol (Section 2.11) reduces forward model error by eliminating $\varepsilon_{\text{unmod}}$. (5) Tier-lifted reconstruction requires a full-physics solver (FDTD, Monte Carlo, P_N) per modality; the current implementation includes P_3 for DOT and FDTD for coherent imaging. Future work: add real-data validation for MRI and ultrasound, extend the tier-lifting solver library to all 22 application domains, improve Performance Agent calibration for scattering systems, and build a community benchmark for imaging system design.

4Methods

4.1Proof of Theorem 2

Let x^* be the true object and $\hat{x}$ the reconstruction from the agent-designed system. Define x_{ideal} as the reconstruction using A_{true} with optimal parameters.

Step 1 (Model approximation). $\|A_{\text{agent}} - A_{\text{true}}\| \leq \varepsilon_{\text{FPB}} + \varepsilon_{\text{spec}} + \varepsilon_{\text{trans}} + \varepsilon_{\text{param}} + \varepsilon_{\text{unmod}}$ by the chain of approximations: true physics $\to$ Tier-2 model ($\varepsilon_{\text{unmod}}$) $\to$ FPB decomposition (ε_{FPB}) $\to$ specification ($\varepsilon_{\text{spec}}$) $\to$ translated DAG ($\varepsilon_{\text{trans}}$) $\to$ parameterized operator ($\varepsilon_{\text{param}}$).

Step 2 (Reconstruction stability). For a regularized reconstruction $\hat{x} = \arg\min_x \|A_{\text{agent}}(x) - y\|^2 + \lambda\,\text{reg}(x)$, the perturbation bound gives:

$$\|\hat{x} - x_{\text{ideal}}\| \leq \frac{1}{\alpha + \lambda} \|(A_{\text{agent}} - A_{\text{true}})(x^*)\|$$

where α is the coercivity constant of $A^T A$ and λ is the regularization strength.

Step 3 (Composition). $\|\hat{x} - x^*\| \leq \|\hat{x} - x_{\text{ideal}}\| + \|x_{\text{ideal}} - x^*\|$. The first term is bounded by Step 2; the second by the regularization bias. The five ε terms propagate multiplicatively through the Lipschitz constants of each stage.

Step 4 (Bounding each term). $\varepsilon_{\text{FPB}} < 0.01$ by the FPB theorem [2]. $\varepsilon_{\text{spec}} = 0$ when all 6 compiler gates pass. $\varepsilon_{\text{trans}} = 0$ when canonical chain matches the registry. $\varepsilon_{\text{param}} \leq L_A \cdot \|\theta_{\text{agent}} - \theta_{\text{true}}\|$ where L_A is computed by the compiler's Lipschitz analysis of nonlinear primitives. $\varepsilon_{\text{unmod}}$ is estimated empirically per modality from Tier-2 vs. full-physics comparison.

4.26-Gate Compiler Implementation

Gate 1 (DAG). The agent's element list is compiled into an `OperatorGraphSpec` (typed Pydantic v2 model). DAG acyclicity is verified via Kahn's algorithm in $O(V + E)$.

Gate 2 (Chain). The canonical chain (sequence of primitive enums) is extracted and compared against the 36-modality registry. Match is exact or within known equivalences (e.g., Σ merged into D).

Gate 3 (Bounds). Node count $N \le 20$, DAG depth $D \le 10$.

Gate 4 (Nonlinear). For each nonlinear node (D, R, Λ): family enum resolved, parameter count ≤ 2, values within physics-based bounds, Lipschitz constant computed.

Gate 5 (Adjoint). Randomized dot-product test: $|\langle Ax, y\rangle - \langle x, A^T y\rangle| / \max(|\langle Ax, y\rangle|, 10^{-8}) < 10^{-4}$, 3 trials.

Gate 6 (Error). When reference A_{true} is available: $\varepsilon = \mathbb{E}[\|A_{\text{agent}}(x) - A_{\text{true}}(x)\| \,/\, \|A_{\text{true}}(x)\|]$ over 5 random non-negative test vectors, threshold < 0.01.

4.3Recovery Ratio Protocol

Four scenarios measure model mismatch impact:

$$\text{I (oracle):} \quad y = A_{\text{true}}(x) + n; \quad \hat{x}_{\text{I}} = \arg\min_x \|A_{\text{true}}(x) - y\|^2 + \lambda\,\text{reg}(x) \tag{2}$$

$$\text{II (mismatch):} \quad y = A_{\text{true}}(x) + n; \quad \hat{x}_{\text{II}} = \arg\min_x \|A_{\text{agent}}(x) - y\|^2 + \lambda\,\text{reg}(x) \tag{3}$$

$$\text{III (auto-correct):} \quad \hat{x}_{\text{III}} = \text{refine}(\hat{x}_{\text{II}}) \text{ with data-consistency} \tag{4}$$

$$\text{IV (calibrated):} \quad \hat{x}_{\text{IV}} = \arg\min_x \left\|A^{\text{cal}}_{\text{agent}}(x) - y\right\|^2 + \lambda\,\text{reg}(x) \tag{5}$$

The recovery ratio $\rho_{\text{recov}} = (\text{PSNR}_{\text{III}} - \text{PSNR}_{\text{II}})/(\text{PSNR}_{\text{I}} - \text{PSNR}_{\text{II}})$ measures auto-correction effectiveness.

4.4Implementation

The framework is implemented in Python (Physics World Model platform). Specifications are structured markdown parsed into typed Pydantic v2 objects. The primitive library contains 101 implementations covering all 11 canonical types across 4 fidelity tiers: Tier-0 (geometric), Tier-1 (Fourier/convolution), Tier-2 (full-wave: FDTD, BPM, Mie), and Tier-3 (learned surrogates). Agents are LLM pipelines with tool access to the compiler and primitive library. Mean compilation time: 0.35 ms. The tier-lifting protocol is implemented as a generic `TierLiftingProtocol` class accepting any pair of Tier-2/Tier-3 forward models, with built-in DOT (P_1/P_3 diffusion) and coherent imaging (angular spectrum/FDTD) corrections. The 36-modality registry, compiler, tier-lifting module, and test suite are publicly available. Source code: https://github.com/integritynoble/Physics_World_Model.

Data Availability

The LoDoPaB-CT dataset is publicly available [3]. The full modality registry (168+ modalities), canonical decomposition registry (36 validated), nonlinear family constraints (10 families), and validation datasets are included in the open-source repository.

Code Availability

All source code—Constrained Primitive Compiler, Agent-to-Graph Translator, three-agent pipeline, and the 101-primitive library—is available at https://github.com/integritynoble/Physics_World_Model under the MIT licence.

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